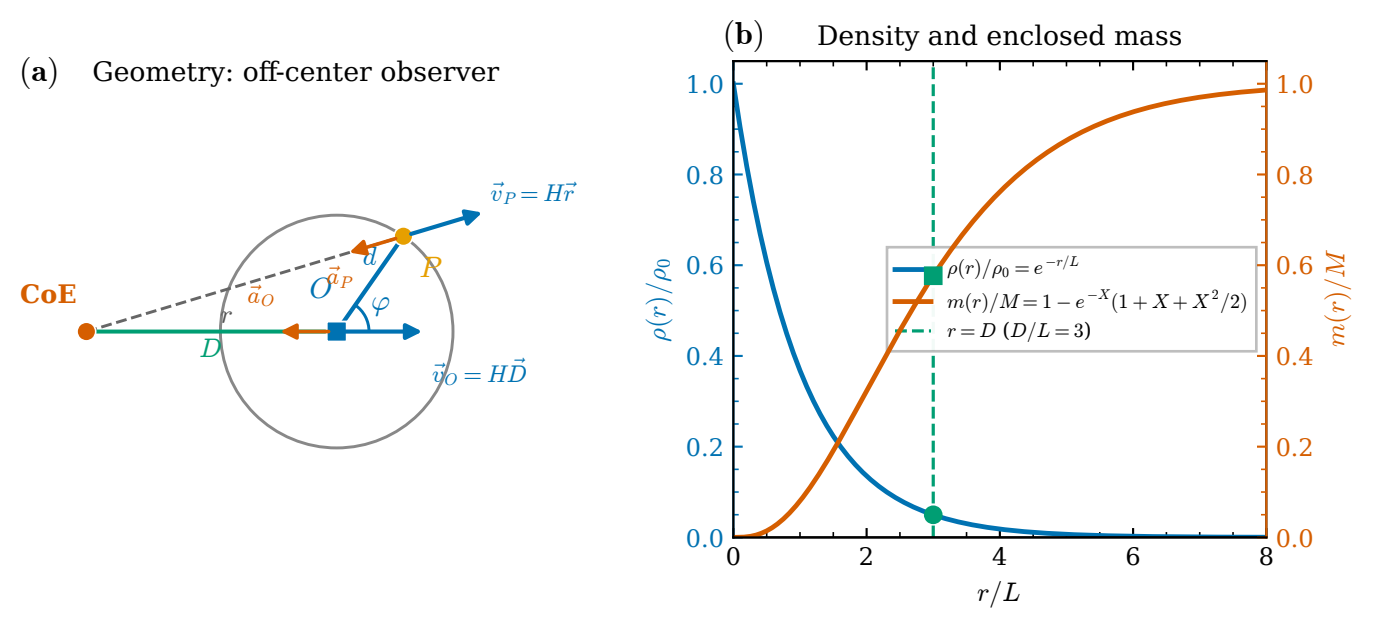
(a) Geometry: off-center observer
CoE
D
r
$\vec{a}_O$
O
$\vec{a}_P$
d
φ
P
$\vec{v}_P = H\vec{r}$
$\vec{v}_O = H\vec{D}$
(b) Density and enclosed mass
$\rho(r)/\rho_0 = e^{-r/L}$
$m(r)/M = 1 - e^{-X}(1 + X + X^2/2)$
$r = D$ $(D/L = 3)$
$\rho(r)/\rho_0$
$m(r)/M$
r/L
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8